

## Lewis Structures (Kelter) — step 0 for (1–8)

- Sum valence e<sup>-</sup> (= **group #**; add 1 per -, remove 1 per + for ions)
- Sum "wants" (table; period 3+ central: see expanded octet below)
- Bonds = (wants - valence) ÷ 2 (round down)
- Central atom = lowest EN (**never H**); ions go in brackets, charge outside
- Remaining e<sup>-</sup> = valence - 2 × bonds: outer atoms first, then central; double/triple bond if central lacks octet or nonzero FC
- FC = valence - (lone e<sup>-</sup> + # bonds): want overall 0, else - on most EN atom

|                    | H | Be | B, Al | C, Si | N, P | O, S | F-I | Noble |
|--------------------|---|----|-------|-------|------|------|-----|-------|
| Val e <sup>-</sup> | 1 | 2  | 3     | 4     | 5    | 6    | 7   | 8     |
| Wants              | 2 | 4  | 6     | 8     | 8    | 8    | 8   | 8     |

- Expanded octet** — period 3+ central (P, S, Cl, Br, I, Xe) can hold >8, so step 3 can undercount bonds. If bonds < # outer atoms: ignore step 3 — single-bond **every** outer atom, complete outer octets, then put **all leftover e<sup>-</sup> on the central** as lone pairs
- ClF<sub>3</sub>: 28 e<sup>-</sup> - 3 bonds (6) - 9 F LP (18) = 4 left → **2 LP on Cl** → 5 groups → T-shaped. Likewise SF<sub>4</sub> (1 LP, seesaw); XeF<sub>4</sub> (2 LP, sq. planar); XeF<sub>2</sub>, I<sub>3</sub><sup>-</sup> (3 LP, linear); PCl<sub>5</sub>, SF<sub>6</sub> (0 LP)

## Periodic Trends

|                        | Trend  | Top | Bot | Left | Right |
|------------------------|--------|-----|-----|------|-------|
| Atomic size (radius)   | ✓ (Cs) | ↓   | ↑   | ↑    | ↓     |
| EN (electronegativity) | ↗ (F)  | ↑   | ↓   | ↓    | ↑     |
| IE (ionization energy) | ↗ (He) | ↑   | ↓   | ↓    | ↑     |
| EA (electron affinity) | ↗ (Cl) | ↑   | ↓   | ↓    | ↑     |

- Cation < neutral < anion; ↑ΔEN → ↑ bond dipole (3–4); smaller ion, higher charge → ↑ΔH<sub>hydr</sub> (18); larger atom → ↑ dispersion (9–12)

| Trend                                     | Bond  | Enthalpy           | Exception                          |
|-------------------------------------------|-------|--------------------|------------------------------------|
| EA (X + e <sup>-</sup> → X <sup>-</sup> ) | form  | <b>exothermic</b>  | endo: Be, Mg, noble, EA2 (usually) |
| IE                                        | break | <b>endothermic</b> | —                                  |

- Lattice energy** U ∝ Q<sub>1</sub>Q<sub>2</sub> / (r<sub>1</sub> + r<sub>2</sub>): heat **released** (exo.) forming ionic solid from gaseous ions (*dissociation* = endo); max U = ↑ charges, ↓ radii; used in ΔH<sub>soln</sub> (18)

## (1–2) Molecular Geometry (VSEPR)

- Central atom** BG + LP (double/triple bond or radical e<sup>-</sup> = 1 group); lone pairs compress angles (SCI2 a little < 109.5°); compound examples in the (3–4) table

| EG | e- geo.     | Hyb.                           | BG | LP | Mol. geo.   | Angle    | Polarity |
|----|-------------|--------------------------------|----|----|-------------|----------|----------|
| 2  | linear      | sp                             | 2  | 0  | linear      | 180°     | nonpolar |
| 3  | tri. planar | sp <sup>2</sup>                | 3  | 0  | tri. planar | 120°     | nonpolar |
| 3  | tri. planar | sp <sup>2</sup>                | 2  | 1  | bent        | <120°    | polar    |
| 4  | tetra.      | sp <sup>3</sup>                | 4  | 0  | tetra.      | 109.5°   | nonpolar |
| 4  | tetra.      | sp <sup>3</sup>                | 3  | 1  | tri. pyr.   | <109.5°  | polar    |
| 4  | tetra.      | sp <sup>3</sup>                | 2  | 2  | bent        | <109.5°  | polar    |
| 5  | tri. bipy.  | sp <sup>3</sup> d              | 5  | 0  | tri. bipy.  | 90/120°  | nonpolar |
| 5  | tri. bipy.  | sp <sup>3</sup> d              | 4  | 1  | seesaw      | 90/120°  | polar    |
| 5  | tri. bipy.  | sp <sup>3</sup> d              | 3  | 2  | T-shaped    | 90, 180° | polar    |
| 5  | tri. bipy.  | sp <sup>3</sup> d              | 2  | 3  | linear      | 180°     | nonpolar |
| 6  | octa.       | sp <sup>3</sup> d <sup>2</sup> | 6  | 0  | octa.       | 90°      | nonpolar |
| 6  | octa.       | sp <sup>3</sup> d <sup>2</sup> | 5  | 1  | sq. pyr.    | 90°      | polar    |
| 6  | octa.       | sp <sup>3</sup> d <sup>2</sup> | 4  | 2  | sq. planar  | 90°      | nonpolar |

## (3–4) Polarity & Dipole Moment

| Compounds                                                                                                            | BG | LP | Shape       | Polarity       |
|----------------------------------------------------------------------------------------------------------------------|----|----|-------------|----------------|
| diatomic: H <sub>2</sub> , Br <sub>2</sub> (same); CO, IBr, HCl (diff.)                                              | 1  | —  | linear      | nonpol. / pol. |
| BeCl <sub>2</sub> , CO <sub>2</sub> , CS <sub>2</sub>                                                                | 2  | 0  | linear      | nonpolar       |
| BF <sub>3</sub> , BH <sub>3</sub> , SO <sub>3</sub>                                                                  | 3  | 0  | tri. planar | nonpolar       |
| SO <sub>2</sub> , O <sub>3</sub> , ClNO                                                                              | 2  | 1  | bent        | polar          |
| CH <sub>4</sub> , CCl <sub>4</sub> , CF <sub>4</sub> , SiH <sub>4</sub> , PO <sub>4</sub> <sup>3-</sup> , hydrocarb. | 4  | 0  | tetra.      | nonpolar       |
| CH <sub>3</sub> Cl/F/Br/I, CH <sub>2</sub> Cl <sub>2</sub> , CHCl <sub>3</sub> (mixed)                               | 4  | 0  | tetra.      | polar          |
| NH <sub>3</sub> , PH <sub>3</sub> , AsH <sub>3</sub> , PCl <sub>3</sub> , NCl <sub>3</sub> , NF <sub>3</sub>         | 3  | 1  | tri. pyr.   | polar          |
| H <sub>2</sub> O, OF <sub>2</sub> , H <sub>2</sub> S, SCI <sub>2</sub> , NH <sub>2</sub> <sup>-</sup>                | 2  | 2  | bent        | polar          |
| PCl <sub>5</sub> , PBr <sub>5</sub>                                                                                  | 5  | 0  | tri. bipy.  | nonpolar       |
| ClF <sub>3</sub>                                                                                                     | 3  | 2  | T-shaped    | polar          |
| XeF <sub>2</sub> , I <sub>3</sub> <sup>-</sup>                                                                       | 2  | 3  | linear      | nonpolar       |
| BrF <sub>5</sub>                                                                                                     | 5  | 1  | sq. pyr.    | polar          |
| XeF <sub>4</sub>                                                                                                     | 4  | 2  | sq. planar  | nonpolar       |

- Dipole** = separation of + / - charge
- (μ ≠ 0) = **polar** = bond dipoles (ΔEN) **don't cancel** (lone pairs)
- CH<sub>3</sub>F > CH<sub>3</sub>Cl > CH<sub>3</sub>Br > CH<sub>3</sub>I; NH<sub>3</sub> > PH<sub>3</sub> > AsH<sub>3</sub> (EN)
- Polar dissolves polar: NH<sub>3</sub> in H<sub>2</sub>O ✓, H<sub>2</sub>S in NH<sub>3</sub> ✓; CCl<sub>4</sub>, SiO<sub>2</sub>, O<sub>2</sub>, CO<sub>2</sub> ✗
- Solvent polarity: H<sub>2</sub>O > ethanol > acetone > CH<sub>2</sub>Cl<sub>2</sub> > diethyl ether > toluene > hexane
- Alkanes C<sub>n</sub>H<sub>2n+2</sub>, all **nonpolar** — C count: meth 1, eth 2, prop 3, but 4, pent 5, hex 6, hept 7, oct 8, non 9, dec 10

## (5–6) Hybridization of an Atom in a Structure

- Look up the EG **of that atom** in the (1–2) table: 2 → sp, 3 → sp<sup>2</sup>, etc.
- C, 4 single bonds → **sp<sup>3</sup>**; C with 1 double bond (C=O, C=C, benzene ring) → **sp<sup>2</sup>**; C with triple or 2 doubles → **sp**
- O in -O-H or ether (2 atoms + 2 LP) → **sp<sup>3</sup>**; carbonyl =O → **sp<sup>2</sup>**
- N with 3 atoms + 1 LP (amine) → **sp<sup>3</sup>**
- Each atom separately: CH<sub>3</sub>-CHO → left C sp<sup>3</sup>, carbonyl C sp<sup>2</sup>

## (7–8) σ and π Bonds: Counting & Concepts

- single** = 1σ; **double** = 1σ + 1π; **triple** = 1σ + 2π
- σ = hybrid orbitals, head-on along the bond axis; π = unhybridized p-p, above & below the axis
- cis** = same side, **trans** = opposite sides; **π restricts rotation** (double/triple bonds)
- CH<sub>3</sub>-CH=CH-COOH: 11σ, 2π; benzene C<sub>6</sub>H<sub>6</sub>: 12σ (6C-C + 6C-H) + 3π
- Two 1s (or head-on p+p along bond axis) → σ and σ\*; sideways p+p (perpendicular axis) → π and π\*
- Benzene delocalized π MOs: e<sup>-</sup> **free to move around the ring**

## (9–10) Intermolecular Forces

| Force                                                     | Found in                                                                |
|-----------------------------------------------------------|-------------------------------------------------------------------------|
| <b>Dispersion</b> (London, van der Waals, induced dipole) | <b>ALL</b> molecules; ↑ w/ size & molar mass; straight-chain > branched |
| <b>Dipole-dipole</b>                                      | polar molecules                                                         |
| <b>H-bond</b>                                             | H bonded <b>directly to N, O, F</b>                                     |
| <b>Ion-dipole</b>                                         | ion + polar solvent (NaCl in H <sub>2</sub> O)                          |

| Compounds                                                                                                                                                                                                                    | Disp. | Dip.-dip. | H-bond |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|--------|
| Br <sub>2</sub> , Cl <sub>2</sub> , F <sub>2</sub> , H <sub>2</sub> , CO <sub>2</sub> , BCl <sub>3</sub> , CF <sub>4</sub> , hydrocarb., noble gases                                                                         | ✓     | ✗         | ✗      |
| HCl, HBr, SO <sub>2</sub> , CHCl <sub>3</sub> , CH <sub>3</sub> Br, CH <sub>3</sub> CHO, CH <sub>3</sub> CN, ethers (CH <sub>3</sub> OCH <sub>3</sub> ), H <sub>2</sub> S, CH <sub>3</sub> SCH <sub>3</sub>                  | ✓     | ✓         | ✗      |
| H <sub>2</sub> O, NH <sub>3</sub> , HF, CH <sub>3</sub> OH, CH <sub>3</sub> CH <sub>2</sub> OH, CH <sub>3</sub> NH <sub>2</sub> , CH <sub>3</sub> COOH, H <sub>2</sub> O <sub>2</sub> , HOCH <sub>2</sub> CH <sub>2</sub> OH | ✓     | ✓         | ✓      |

## (11–12) Rank Vapor Pressure & Boiling Point

- Ionic > H-bond > dipole-dipole > dispersion** (at similar size)

| IMF | bp | mp | Visc. | Surf. tens. | ΔH <sub>vap</sub> | Crit. T | VP |
|-----|----|----|-------|-------------|-------------------|---------|----|
| ↑   | ↑  | ↑  | ↑     | ↑           | ↑                 | ↑       | ↓  |
| ↓   | ↓  | ↓  | ↓     | ↓           | ↓                 | ↓       | ↑  |

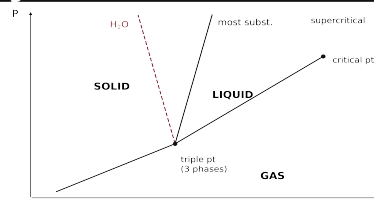
| Rank            | Example                                                                                                                                               |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| VP ↓            | CF <sub>4</sub> > PF <sub>5</sub> > BrF <sub>3</sub> (polar = lowest VP)                                                                              |
| bp ↑            | CH <sub>4</sub> < CH <sub>3</sub> Cl < CH <sub>3</sub> OH < RbCl (ionic)                                                                              |
| bp (disp. size) | I <sub>2</sub> > Br <sub>2</sub> > Cl <sub>2</sub> > F <sub>2</sub> ; CH <sub>3</sub> CHO > CH <sub>3</sub> OCH <sub>3</sub> > Rn                     |
| bp (H-bond)     | HF > HI > HBr > HCl                                                                                                                                   |
| bp (# of OH)    | HOCH <sub>2</sub> CH <sub>2</sub> OH > ClCH <sub>2</sub> CH <sub>2</sub> OH > CH <sub>3</sub> CH <sub>2</sub> OH > CH <sub>3</sub> CH <sub>2</sub> Cl |

## (13) Viscosity, Capillary Action, Surface Tension

|                         |                                                                                                          |
|-------------------------|----------------------------------------------------------------------------------------------------------|
| <b>Surface tension</b>  | energy required to stretch/increase surface <b>area</b> by unit amount                                   |
| <b>Viscosity</b>        | resistance of a liquid to <b>flow</b> ; ↑ with IMF, ↓ with ↑T; spherical/compact → lower                 |
| <b>Capillary action</b> | liquid flows <b>up a thin tube against gravity</b> = cohesion (like molecules) + adhesion (liquid-glass) |

- Concave meniscus (H<sub>2</sub>O): adhesion > cohesion; convex (Hg): cohesion > adhesion. Highest viscosity = most H-bonding (HOCH<sub>2</sub>CH<sub>2</sub>OH)

## (14) Phase Diagrams



- Lines = 2 phases coexist; **triple point** = all 3; **critical point**: above T<sub>c</sub> no pressure can liquefy gas → **supercritical fluid**
- ↑T at const P (→): solid → liquid → gas. ↑P at const T (↑): gas → liquid → solid. Normal bp/mp read at 1 atm

| Endothermic (+ΔH, +ΔS) | Exothermic (-ΔH, -ΔS) |
|------------------------|-----------------------|
| melting s→l            | freezing l→s          |
| vaporization l→g       | condensation g→l      |
| sublimation s→g        | deposition g→s        |

- Order: solid = greatest, gas = least (↑T = ↓ order)
- H<sub>2</sub>O**: fusion line leans **left** (ice less dense) → ↑P melts ice. **CO<sub>2</sub>**: triple pt at 5.1 atm → at 1 atm solid **sublimes** s→g (dry ice)

## (15–16) Type of Solid

| Type                         | Held by                   | Properties                         | Examples                                                         |
|------------------------------|---------------------------|------------------------------------|------------------------------------------------------------------|
| <b>Ionic</b>                 | ± attraction              | hard, brittle, high mp, poor cond. | NaCl, AgCl, CaF <sub>2</sub> , MgO, CaCO <sub>3</sub>            |
| <b>Molecular</b>             | IMFs                      | soft, low mp, poor cond.           | ice H <sub>2</sub> O, CO <sub>2</sub> , I <sub>2</sub> , sucrose |
| <b>Metallic (atomic)</b>     | e- sea                    | low-high mp, <b>good cond.</b>     | Au, Cu, Fe, Na                                                   |
| <b>Network cov. (atomic)</b> | covalent bonds            | v. hard, v. high mp, poor cond.    | diamond, quartz SiO <sub>2</sub> , graphite, SiC                 |
| <b>Nonbonding (atomic)</b>   | dispersion                | v. low mp                          | noble gases: Ar(s)                                               |
| <b>Amorphous</b>             | no long-range order       | melts over a <b>range</b>          | glass, rubber, plastic, butter, wax                              |
| <b>Ceramic</b>               | inorganic nonmetal, fired | brittle, heat-resistant            | clay/kaolinite, porcelain, Al <sub>2</sub> O <sub>3</sub>        |

- Graphite sp<sup>2</sup> sheets: conducts along the sheets, slippery; diamond sp<sup>3</sup>: insulator
- FCC packing = most space-efficient (12 neighbors)

### (17) Band Theory & Doping

- Bands = MOs delocalized over the whole crystal: **valence band = bonding MOs, conduction band = antibonding MOs, band gap** between
- Conductor** (metal): bands continuous/overlap; **Semiconductor** (metalloids Si, Ge): small gap, e- promotable; **Insulator: LARGE gap**
- Doping** = add impurity to ↑ conductivity:

|              | Dopant                              | Carrier                                                    |
|--------------|-------------------------------------|------------------------------------------------------------|
| n-type       | Group <b>V/5A</b> (P, As)           | extra e- in <b>conduction band</b> (– charges)             |
| p-type       | Group <b>III/3A</b> (B, Al, Ga, In) | <b>holes</b> in <b>valence band</b> (e- hop between holes) |
| p-n junction | —                                   | <b>diode</b> : current flows one way                       |

### (18) Heat of Solution & Hydration

$$\Delta H_{\text{solution}} = \Delta H_{\text{hydration}} - \Delta H_{\text{lattice energy}}$$
$$\Delta H_{\text{solution}} = \Delta H_{\text{solute}} + (\Delta H_{\text{solvent}} + \Delta H_{\text{mix}})$$
$$= (-\Delta H_{\text{lattice}}) + \Delta H_{\text{hydration}}$$

- Dissolving = break crystal (**ΔHsolute = –ΔHlattice**, costs energy) + make ion-dipole attractions (**ΔHsolvent + ΔHmix = ΔHhydration**, releases)
- ΔHhydration**: heat **released** when 1 mol **gaseous ions** dissolves in water (always exo)
- Highest ΔHhydr**: smallest ion + highest charge: Al3+ > Mg2+ > Na+ > K+; F- > Cl-
- ΔHsoln **exo (–)**: ΔHhydration > ΔHsolute; **endo (+)**: solute > hydration; **≈ 0**: comparable

### (19) Saturation; Like Dissolves Like

|                       |                                                                                                |
|-----------------------|------------------------------------------------------------------------------------------------|
| <b>Unsaturated</b>    | less than equilibrium amount — more can dissolve                                               |
| <b>Saturated</b>      | max amount at that T; dissolved ⇌ solid <b>dynamic equilibrium</b>                             |
| <b>Supersaturated</b> | <b>more</b> than equilibrium amount; unstable — seed crystal/scratch → excess crystallizes out |

- Solution forms if solute-solvent attraction **comparable** to solute-solute + solvent-solvent; polar/ionic ⇌ polar (H2O, ethanol, NH3); nonpolar ⇌ nonpolar (hexane, CCl4, benzene)
- Most soluble in ethanol = most H-bonds (ethylene glycol); in hexane = longest chain (1-pentanol); all **nitrates** soluble

### (20) Solubility vs P & T; When Bubbles/Boiling

| ↑ gives              | Gas                  | Solid / liquid |
|----------------------|----------------------|----------------|
| ↑ <b>Pressure</b>    | ↑ solubility (Henry) | no effect      |
| ↑ <b>Temperature</b> | ↓ solubility         | ↑ solubility   |

- Dissolve the most gas: **low T + high P** (cold soda holds CO2)
- Liquid **boils** when its vapor pressure = **external (atm) pressure**; gas **bubbles out** of solution when its partial (vapor) pressure > **atm pressure**

### (21) Molarity (and Molality)

$$M = \frac{\text{mol solute}}{L \text{ solution}} \quad m = \frac{\text{mol solute}}{\text{kg solvent}}$$

- mL ÷ 1000 = L; g ⇌ mol via molar mass; dilution: M1V1 = M2V2
- mol = M × V(L): 0.500 M × 0.250 L = **0.125 mol**
- V = mol ÷ M: 0.300 mol ÷ 0.450 M = 0.667 L = **667 mL**
- Molality m uses **kg of solvent**, not solution — for ΔTf/ΔTb
- Only molarity changes with temperature** (volume swells); molality, mass %, χ don't (mass/count-based)

### (22) Percent Concentration (% w/v)

$$\% w/v = \frac{g \text{ solute}}{100 \text{ mL soln}} \Rightarrow g = \frac{\% \times \text{mL}}{100}$$

- g = % × mL ÷ 100: for 250 mL of 1.5% → 1.5 × 250 ÷ 100 = **3.75 g** (the summary's "31.25 g" contradicts its own formula)
- mL = g ÷ (% ÷ 100): 16.6 g ÷ 0.0450 g/mL = **369 mL**
- % m/m** = g solute ÷ (g solute + g solvent) × 100 — denominator = **whole solution**
- g water = g solution × (1 – % ÷ 100): 300 g of 9.71% KBr → 300 × 0.9029 = **271 g water**
- % v/v** = vol component ÷ total vol × 100

### (23) Henry's Law (Cgas = kH·Pgases)

$$C_{\text{gas}} = k_H P_{\text{gas}} \quad k_H = C_{\text{gas}} / P_{\text{gas}}$$

- kH = "solubility constant of a gas" (M/atm); C = molar solubility
- kH = C ÷ P** (divide — below 1 atm, kH comes out **larger** than C)
- 1 atm** = 760 mmHg = 760 torr = 14.7 psi = 101.3 kPa (mmHg → atm: ÷ 760)

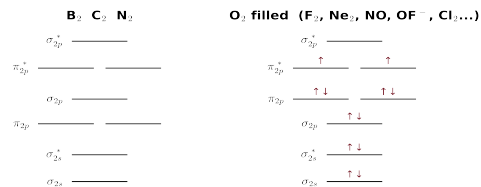
Given C = 4.7e–4 mol/L at 522 mmHg:

- P = 522 ÷ 760 = 0.687 atm
- kH = 4.7e–4 ÷ 0.687 = **6.8e–4 mol/L·atm**

### (24) Osmosis

- Osmosis: solvent** (water) flows across a semipermeable membrane (passes solvent, **blocks solute** ions — it does *not* stop water) from **low solute conc.** → **high solute conc.**
- Hyperosmotic** = more solute: water flows **into** it (cell shrinks)
- Hyposmotic** = less solute: water flows **out** to the more concentrated side (cell swells)
- Isosmotic** = equal: no net flow (isotonic saline = 0.9% w/v NaCl); n = pressure needed to **stop** osmotic flow

### (25) MO Theory — Bond Order & Magnetism

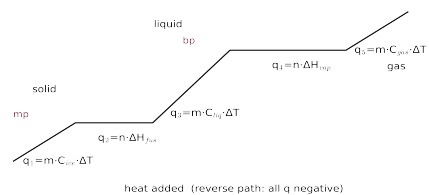


- Valence e- total: **add 1 per – charge, remove 1 per +**
- Fill low → high, 2 per orbital, Hund across the n pair
- BO = (bonding e- – antibonding e-) / 2**
- Any unpaired e- → **paramagnetic**; all paired → **diamagnetic**
- BO > 0 → stable, can form (H2 1 ✓; He2 0 x can't exist); **higher BO = stronger & shorter bond** (N2+ weaker/longer than N2). σ2s + σ\*2s cancel, so counting 2p alone gives the same BO

| Sp.     | e- | BO  | Mag  | Sp.  | e- | BO  | Mag              |
|---------|----|-----|------|------|----|-----|------------------|
| N2      | 10 | 3   | dia  | O2   | 12 | 2   | <b>para</b> (2↑) |
| N2+     | 9  | 2.5 | para | O2+  | 11 | 2.5 | para             |
| N2-     | 11 | 2.5 | para | O2-  | 13 | 1.5 | para             |
| F2, Cl2 | 14 | 1   | dia  | Cl2+ | 13 | 1.5 | para             |
| B2      | 6  | 1   | para | Cl2- | 15 | 0.5 | para             |

- Heteronuclear (NO, OF<sup>–</sup>) → the O2-type (right) ordering (more EN atom sits lower); period 3 (Cl2, Na2) → same shapes, n = 3

### (26) Phase-Change Heat (gas ⇌ solid)



- Sketch the path; skip segments you don't cross (see examples below)
- Sloped (one phase): **q = m(g)·Cs·ΔT**; flat (phase change): **q = n(mol)·ΔH**
- Sum; convert J → kJ. Cooling path (gas → solid): every q **negative** (ΔHcond = –ΔHvap, ΔHfreeze = –ΔHfus)
- H2O: C(ice) 2.09, C(liq) 4.184, C(steam) ≈ 2.0 J/g·°C; ΔHfus 6.02, ΔHvap 40.7 kJ/mol; MM 18.02
- 36.0 g H2O, liquid 65 °C → gas 115 °C:
  - heat liquid 65→100: q = 36.0 g × 4.184 × 35° = 5.27 kJ
  - boil (use mol): q = (36.0 ÷ 18.02) × 40.7 = 81.3 kJ
  - heat gas 100→115: q = 36.0 g × 2.01 × 15° = 1.09 kJ
  - total = **87.7 kJ**
- Cooling runs the same steps backward: 125 g benzene (MM 78.11), gas 425 K → liquid 335 K (bp 353, mp 279 — it never freezes): 9.5 kJ + 54.3 kJ + 3.9 kJ = **67.7 kJ removed**
- Endo (+): melt, vaporize, sublime; Exo (–): freeze, condense, deposit; ΔHsub = ΔHfus + ΔHvap

### (27) Raoult's Law (VP Lowering)

$$P_{\text{soln}} = \chi_{\text{solvent}} P_{\text{solvent}}^{\circ} \quad \Delta P = \chi_{\text{solute}} P_{\text{solvent}}^{\circ}$$
$$\chi_{\text{solute}} = \frac{\text{mol solute}}{\text{mol solute} + i \cdot \text{mol solute}}$$

- mol solvent = g ÷ MM; mol solute = g ÷ MM (**x i if ionic**)
- χ(solvent) = mol solvent ÷ total mol — always < 1
- P(soln) = χ(solvent) × P°; lowering ΔP = P° – P(soln) = χ(solute)·P°
  - 75.0 g citric acid (192.12) + 420 g H2O, P° 71.93 mmHg: 0.390 & 23.31 mol → χ = 0.9835 → **70.7 mmHg**

### (28) Freezing ↓ / Boiling ↑ (van't Hoff)

$$\Delta T_f = i K_f m \quad \Delta T_b = i K_b m$$

- m = mol solute / **kg solvent**; Tf = T°f – ΔTf, Tb = T°b + ΔTb
- i** = particles per formula unit
- Lowest fp for equal grams = small MM and high i (NaOH), **but the HW answer key rewards the most ions per formula (CaCl2, i = 3)**
- Molar mass from fp — 20.0 g unknown per 500 g benzene (T°f 5.444, Kf 5.12) freezes at 3.77 °C:
  - ΔTf = 5.444 – 3.77 = 1.674 °C
  - m = ΔTf ÷ (i × Kf) = 1.674 ÷ 5.12 = 0.327 mol/kg
  - mol = m × kg solvent = 0.327 × 0.500 = 0.163 mol
  - MM = g ÷ mol = 20.0 ÷ 0.163 = **≈120 g/mol**

### (29) Osmotic Pressure → Molar Mass

$$\pi = i M R T \Rightarrow M = \frac{\pi}{i R T}$$

- R = **0.08206** L·atm/mol·K; T in **K** (+273.15); n in **atm** (torr ÷ 760); i = 1 nonelectrolyte
- Given 15.2 **mg** in 150.0 mL, n = 8.44 torr, 25 °C:
  - M = n ÷ (iRT) = (8.44 ÷ 760) ÷ (0.08206 × 298) = 4.54e–4 M
  - mol = M × V(L) = 4.54e–4 × 0.1500 = 6.81e–5 mol
  - MM = g ÷ mol = 0.0152 ÷ 6.81e–5 = **223 g/mol**